

VARFU OMU, Romania

WMO: 152800

Lat: 45.4460N

Lon: 25.4563E

Elev: 8223

StdP: 10.82

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-3.7	-19.9	2.5	8.2	-13.2	3.7	6.9	76.8	14.2	76.2	13.6	27.3	320	1.082

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	58.4	50.6	55.6	48.8	53.2	47.6	52.6	55.9	50.6	53.7	49.0	51.8	8.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
51.2	76.2	54.1	49.3	70.8	52.0	47.6	66.5	50.2	25.1	56.2	23.7	53.8	22.6	51.8	60.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 75.0	54.5	45.1	-12.9	63.3	4.1	3.4	-15.9	65.7	-18.3	67.7	-20.6	69.6	-23.6	72.1		
(5)			-13.2	56.3	4.1	2.3	-16.1	58.0	-18.5	59.3	-20.8	60.6	-23.7	62.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.2	14.0	14.6	17.5	25.7	34.3	41.0	44.6	45.3	37.9	31.8	24.8	17.3		
	DBStd	13.56	9.05	8.71	8.13	6.88	6.26	6.31	5.73	5.60	6.99	8.61	9.36	9.02		
	HDD50	7654	1116	992	1008	728	485	274	185	165	366	566	755	1014		
	HDD65	13081	1581	1412	1473	1178	950	719	633	609	812	1030	1205	1479		
	CDD50	48	0	0	0	0	0	6	17	21	3	1	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	18.0	22.7	22.3	21.4	17.6	16.7	14.6	13.7	12.1	15.9	16.7	21.4	21.5		
	PrecAvg	34.8	1.8	1.7	2.0	2.8	4.1	4.7	4.4	3.8	2.9	2.5	2.1	2.0		
	PrecMax	48.8	4.9	3.3	3.7	5.7	7.7	8.5	7.8	8.1	5.8	5.4	4.2	4.8		
(15) Precipitation	PrecMin	22.6	0.3	0.3	0.5	1.4	1.4	2.0	0.5	1.2	0.8	0.2	0.2	0.3		
	PrecStd	6.5	1.1	0.7	0.9	1.1	1.6	1.8	1.9	1.8	1.4	1.3	1.1	1.1		
	0.4%	DB	34.1	34.8	35.9	43.9	51.3	58.8	61.6	62.0	58.2	53.3	46.1	40.1		
	0.4%	MCWB	27.4	29.4	31.0	35.9	45.3	50.0	52.1	53.0	47.8	42.9	36.2	30.3		
	2%	DB	30.3	30.6	32.8	40.0	47.4	54.9	57.8	58.5	52.9	49.0	42.3	33.5		
(23) Mean Coincident Wet Bulb Temperatures	2%	MCWB	24.9	27.0	29.2	35.3	42.2	48.6	50.2	51.4	44.8	38.3	34.1	28.2		
	5%	DB	27.4	27.5	30.2	36.7	44.8	52.1	55.0	55.5	49.4	45.6	39.2	30.2		
	5%	MCWB	24.6	24.9	28.1	33.2	40.6	47.0	48.9	49.2	42.7	37.1	31.3	26.5		
	10%	DB	25.2	25.5	27.9	34.0	42.4	49.5	52.5	52.8	46.5	42.7	36.2	28.0		
	10%	MCWB	23.2	23.4	26.3	31.9	39.1	45.5	47.9	47.7	41.5	36.2	30.5	25.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.8	31.0	32.9	39.1	46.8	53.5	55.2	56.3	50.1	47.3	40.4	33.2		
	0.4%	MCDB	31.1	33.2	34.3	42.0	49.3	56.2	58.7	59.9	55.2	50.3	43.1	35.5		
	2%	WB	27.1	27.7	30.3	36.0	43.9	50.3	52.3	53.0	46.7	42.2	36.2	29.7		
	2%	MCDB	28.4	29.2	31.5	38.4	46.1	53.4	55.5	56.4	50.4	45.0	38.5	31.7		
	5%	WB	25.1	25.7	28.4	33.9	41.6	48.1	50.4	50.7	44.7	39.8	34.2	27.8		
(31) Mean Daily Temperature Range	5%	MCDB	26.4	27.0	29.7	35.9	43.5	50.8	53.2	53.7	47.6	42.5	36.4	29.2		
	10%	WB	23.3	23.7	26.5	32.2	39.9	46.1	48.8	48.9	42.9	37.8	32.2	26.1		
	10%	MCDB	24.9	25.1	27.7	33.4	41.8	48.5	51.4	51.7	45.3	40.4	34.5	27.7		
	5% DB	MDBR	8.5	8.8	9.4	8.4	7.7	8.1	8.3	8.5	8.1	8.7	8.4	8.5		
	5% DB	MCDBR	9.7	10.0	10.9	10.5	10.5	11.4	11.6	11.3	11.0	10.8	10.4	9.3		
(36) Clear-Sky Solar Irradiance	5% WB	MCWBR	8.5	9.0	9.8	8.9	9.7	10.5	10.7	11.1	10.7	9.9	8.0	7.9		
	5% WB	MCDBR	9.2	9.3	9.9	8.9	9.6	10.3	10.4	10.5	9.7	9.7	8.0	8.2		
	5% WB	MCWBR	8.7	9.0	9.7	8.3	9.3	10.1	10.5	10.8	9.9	10.0	7.8	7.9		
	taub		0.223	0.240	0.266	0.303	0.295	0.297	0.306	0.313	0.287	0.264	0.247	0.225		
	taud		2.467	2.442	2.474	2.449	2.500	2.504	2.470	2.443	2.519	2.588	2.590	2.513		
(42) All-Sky Solar Radiation	Ebn at Noon		295	307	310	305	309	307	303	297	299	294	283	283		
	Edh at Noon		21	26	30	33	33	33	34	33	28	23	19	18		
	RadAvg		444	705	1044	1360	1656	1792	1823	1649	1210	837	490	363		
(45)	RadStd		46	90	109	150	162	163	157	137	117	107	50	44		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.94	N/A	N/A	N/A	N/A	N/A	-329	-349	N/A	N/A
(47)	Regional (69 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	-218	-297	+117	+31

Nomenclature: See separate page